MA2071

Solutions to Midterm Exam

1) (i) We have $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$ or $ax+by+cz+d=0$ where $d=-(ax_0+by_0+cz_0)$.

(ii) True.

(iii) First, note that the planes are parallel as their normal vectors $\langle 10,2,-2\rangle$ and $\langle 5,1,-1\rangle$ are parallel. To find D, choose any point on one plane and calculate the distance from it to the other plane. Putting $y=z=0$ into the first equation yields $10x=5$ and so the point $(1/2,0,0)$ is on this plane. By the lecture notes (or practice midterm), the distance between $(1/2,0,0)$ and the plane $5x+y-z-1=0$ is

$$D=\frac{|5(1/2)+1(0)-1(0)-1|}{\sqrt{5^2+1^2+(-1)^2}}=\frac{\sqrt{3}}{6}.$$

2) (i) See problem 8 (i) on Exercise Set 1.

(ii) See the lecture notes.

3) (i) See problem 4 (i) on the practice midterm.

(ii) Let $F(x,y,z)=yz-\ln(x+z)$. Then $\nabla F(x,y,z)=\left\langle -\frac{1}{x+z},z,y-\frac{1}{x+z}\right\rangle$. So, $\nabla F(0,0,1)=\langle -1,1,-1\rangle$. Thus, we have $-1(x-0)+(1)(y-0)-1(z-1)=0$ or $x-y+z=1$.

4) See the lecture notes.